Mansi Sheth

Security Researcher · Applied Cryptography

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PROFILE

Eighteen years of security research across static analysis, application security, applied cryptography, and vulnerability intelligence have built one consistent instinct: find the hard problem, go deep, and share what you learn. **Three granted patents**, **10+ peer-reviewed conference talks**, and a published technical blog series are the artifacts of that instinct. Now at the **AI × cybersecurity intersection** — where AI is a powerful new toolset for tackling problems that have always been hard, and where it simultaneously creates entirely new ones worth solving. Open to research and product development roles where deep security expertise and AI come together.

HIGHLIGHTS

- **3 Granted Patents** — Static analysis, compiler tooling, software security
- **10+ Conference Talks** — DEF CON · NorthSec · SecTor · JavaOne · AppSec USA · JavaZone · XML Prague
- **Java Crypto Blog Series** — Practical cryptography implementation guides (formerly on Veracode Engineering Blog)

  → *Full details at 1mansis.github.io*

PROFESSIONAL EXPERIENCE

Member of Technical Staff · VulnCheck *2025 – Present*

- Research and maintain ingestion methodologies across 600+ **vulnerability intelligence sources** — spanning CVEs, vendor disclosures, EOL advisories, package manager feeds, threat actor trackers, PoC repositories, and ICS/OT and IoMT ecosystems — delivering **structured, machine-readable** data to security products, SIEMs, VM platforms, and enterprise and government consumers.
- Design and own the data quality standards for cross-source normalization and schema consistency — ensuring the intelligence meets reliability bar that **AI-driven triage, automated detection,** and downstream security tooling depend on.
- Contributed to VulnCheck's **AI roadmap** — provided use case definition and architecture input for the official VulnCheck MCP Server; sole author of VulnCheck Agent Tools, Claude Code skills enabling natural language vulnerability intelligence queries directly inside AI coding assistants.
- Built and deployed an **ML classifier (TF-IDF + Logistic Regression)** to categorize exploit types within VulnCheck's exploit taxonomy — now live on our endpoints serving paying customers; used **LLM-assisted labeling** to handle edge cases and expand training coverage for more robust classification across sparse categories.

Lead Cryptography Software Engineer · SandboxAQ *2023 – 2024*

- Technical Lead for Sandwich: a **crypto-agile, post-quantum-ready** library built to cover all cryptographic primitives — asymmetric encryption, digital signatures, hashing — with swappable classical and PQC backends across C/C++, Python, Go, and Rust; the TLS layer was open-sourced before the project concluded (github.com/sandbox-quantum/sandwich).
- Architected the swappable crypto-backend pattern enabling seamless algorithm transitions — classical to PQC — across C/C++, Python, Go, and Rust bindings; validated the design by integrating Sandwich into internal SandboxAQ products to achieve crypto-agility in practice.
- Led a team of **4 cryptography engineers** — directing technical work, resolving implementation blockers, and keeping the team aligned to product requirements and enterprise adoption goals.

Sr. Principal Security Researcher · Veracode *2013 – 2023*

Security Research

- Primary researcher for the **Java ecosystem** — Veracode's highest-revenue platform — owning security research and detection coverage across the full **CWE and OWASP** vulnerability taxonomy, spanning the JDK, major application and web frameworks, and the vast open-source library landscape that enterprise Java applications depend on.
- Pioneered security research for Python and Go at Veracode — the first to establish vulnerability detection coverage for each language, going deep on ecosystem-specific security anti-patterns, CWE mappings, and detection strategies across their respective framework and library landscapes; developed supporting research tooling to enable scalable analysis.
- Distilled deep Java cryptography research into a language-agnostic blueprint — defining detection logic, flaw categorization, and remediation guidance for crypto vulnerabilities — adopted as the standard specification across all languages and frameworks.
- Formally designated by management as technical mentor for new hires and interns — responsible for onboarding, task assignment, and technical direction within the research team.

Research & Development

- Received **Thought Leadership award** for the highest number of approved patents in company history; patented **3 novel compiler and tooling approaches** for static analysis of non-grammar-based and templating languages.
- Architected a secure research data pipeline — EC2-hosted binary metadata extraction, parsed and stored in a Native XML Database for structured querying, with client-side cryptographic protections and KMS-managed keys for encrypted storage back to S3.
- Leveraged the pipeline for **prevalence analysis** — identifying what frameworks, libraries, and APIs were actually in use across the customer base, allowing research to focus on the highest-impact areas of large ecosystems.
- Applied **ML + NLP** to locate security-sensitive functionality within codebases — enabling research to treat vulnerabilities discovered in those areas with higher investigation priority.

Sr. Technology Analyst, Enterprise Application Security · Fidelity Investments *2007 – 2013*

Penetration Testing

- Conducted security assessments and penetration testing on Fidelity's critical flagship web applications, APIs, and internal infrastructure; led targeted red-team exercises across high-value financial systems.
- Administered and extended AppScan and Burp Suite for enterprise pen testing — writing Burp plugins for cross-tool result aggregation and reporting, and exploring JavaScript automation for session handling within AppScan workflows.

Static Analysis & Tooling

- Lead developer of **'Assembly Line'**: an internal SAST orchestration platform extending commercial static analysis tools with custom Fortify rules, bridging coverage gaps across Fidelity's proprietary frameworks.
- Developed deep expertise in Fidelity's custom application frameworks and Fortify's detection engine — writing custom rules to automate vulnerability discovery across **200+ applications** built on those frameworks.
- Leveraged that framework knowledge to advise security architecture teams on anti-patterns and secure design recommendations — directly influencing how Fidelity's internal frameworks were hardened.

Open Source

- Contributor to OWASP iGoat; authored the Keychain exercise module used in mobile security training.

Lead Software Developer · Intralinks · Accenture · IRIS Ltd *2002 – 2007*

- At Accenture, worked exclusively on JP Morgan's Futures and Options modernization project — and led security remediation of a comprehensive third-party penetration test report, owning end-to-end fixes across a broad range of application security vulnerabilities with a team of two junior engineers. First hands-on introduction to application security.
- At IRIS Ltd, developed dynamic web products for mid-size Indian businesses and contributed a core module to HDFC Bank's platform.

M.S. Computer Science · University of Massachusetts Lowell *2011*

B.E. Computer Engineering · University of Mumbai *2002*